```python
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.widgets import Slider, Button

# Parameter t
t = np.linspace(0, 2*np.pi, 500)

# Initial radius
r0 = 3

# Initial circle
x = r0 * np.cos(t)
y = r0 * np.sin(t)

# Create Figure and Axes
fig, ax = plt.subplots()

# Make space for slider and button
plt.subplots_adjust(bottom=0.30)

# Plot circle
line, = ax.plot(x, y, linewidth=2)

ax.set_xlim(-6, 6)
ax.set_ylim(-6, 6)
ax.set_aspect('equal')

ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title('Interactive Circle: x = r cos(t), y = r sin(t)')
ax.grid(True)

# Slider Axes
slider_ax = fig.add_axes([0.20, 0.12, 0.60, 0.04])

# Create slider
radius_slider = Slider(
    slider_ax,
    'Radius r',
    1,
    5,
    valinit=r0
)

# Function called when slider moves
def update(val):
    r = radius_slider.val

    x = r * np.cos(t)
    y = r * np.sin(t)

    line.set_xdata(x)
    line.set_ydata(y)

    fig.canvas.draw_idle()

# Connect slider to update function
radius_slider.on_changed(update)

# Reset button
button_ax = fig.add_axes([0.40, 0.03, 0.20, 0.05])

reset_button = Button(button_ax, 'Reset')

```

```
65 def reset(event):
66     radius_slider.reset()
67 
68 reset_button.on_clicked(reset)
69 
70 plt.show()
```